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CLOUD COMPUTING

Classroom Handout & MCQ Bank for Accenture and Placement Tests

Concepts • AWS/Azure/GCP Mapping • Security & Networking • 184 MCQs from Basic to Advanced

How to use this handout

Part A covers every cloud concept tested in Accenture's 'Fundamentals of Networking' and 'Computer Science' sections, where cloud, virtualization, containers and DevOps now appear regularly. **Part B** has 184 MCQs in four levels – **L1 Basic**, **L2 Intermediate**, **L3 Accenture-pattern scenarios**, **L4 Advanced** – plus table-based case studies, a second 30-question Accenture scenario set a 20-question placement-paper set (TCS, Infosys, Capgemini, Cognizant, Wipro pattern) and a 22-question recent-drive set on deployment models and TCP connections.

[A] marks questions written on concepts asked in recent Accenture drives (scalable storage, deployment models for sensitive data, hybrid connectivity, VMs vs containers, container isolation, VPN types, latency troubleshooting). **[P]** marks other placement-test patterns. All questions are original Talent Shine questions with verified answers.

Part A – Concept Notes

A1. What is Cloud Computing?

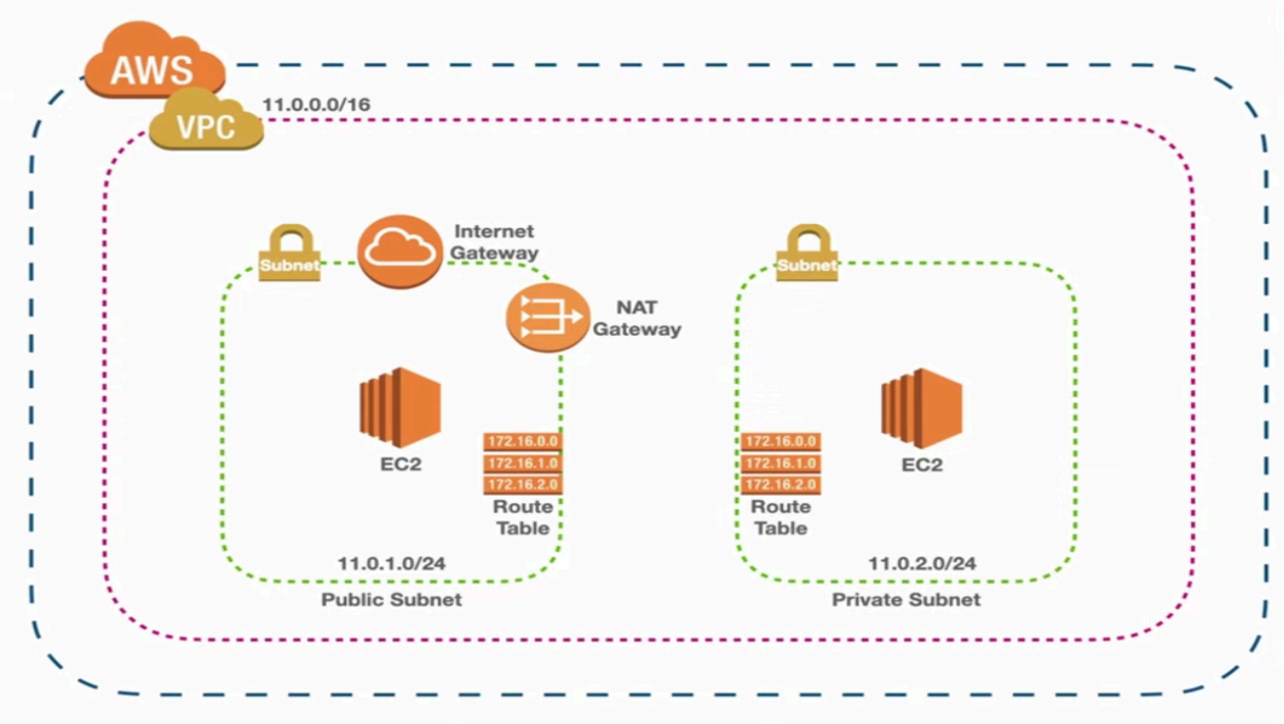
Cloud computing is the on-demand delivery of computing resources – servers, storage, databases, networking, software and analytics – over the internet with pay-as-you-go pricing. NIST defines it by five essential characteristics:

Characteristic	Meaning	Example
On-demand self-service	Provision resources yourself without contacting the provider	Launch a VM from a web console in minutes
Broad network access	Accessible over the network from any device	Use Google Drive from phone and laptop
Resource pooling	Provider serves many customers from shared physical resources (multi-tenancy)	Many companies' VMs on the same data-centre hardware
Rapid elasticity	Scale out and in quickly, often automatically	Add servers during a festival sale, remove them after
Measured service	Usage is metered and billed	Pay per GB stored or per hour of compute

Benefit	Explanation
CapEx → OpEx	No upfront hardware purchase; pay monthly for what is used
Scalability & elasticity	Match capacity to demand instead of over-provisioning

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Benefit	Explanation
Speed & agility	New environments in minutes
Global reach	Deploy close to users in many regions
Reliability	Redundant data centres, backups and disaster recovery options
Focus on business	Provider manages hardware, power, cooling and much of the platform



A2. Service Models

Model	You manage	Provider manages	Examples
IaaS – Infrastructure	OS, runtime, middleware, applications, data	Physical servers, storage, network, virtualization	AWS EC2, Azure Virtual Machines, Google Compute Engine
PaaS – Platform	Applications and data	Everything up to the runtime	Azure App Service, Google App Engine, AWS Elastic Beanstalk, Heroku
SaaS – Software	Your data and user access	The entire application	Gmail, Microsoft 365, Salesforce, Zoom
FaaS – Serverless	Function code	Servers, scaling, runtime	AWS Lambda, Azure Functions, Google Cloud Run functions
CaaS – Containers	Container images and configuration	Container platform and orchestration	Amazon EKS/ECS, Azure AKS, Google GKE

Shared responsibility model

Layer	On-premise	IaaS	PaaS	SaaS
Data & access	Customer	Customer	Customer	Customer
Application	Customer	Customer	Customer	Provider
Runtime & middleware	Customer	Customer	Provider	Provider
Operating system	Customer	Customer	Provider	Provider
Virtualization	Customer	Provider	Provider	Provider
Servers, storage, network, data centre	Customer	Provider	Provider	Provider

Rule to remember: the provider is responsible for security **of** the cloud; the customer is always responsible for security **in** the cloud – data, identities and configuration.

A3. Deployment Models

Model	Description	Best for
Public	Shared infrastructure owned by a provider, accessed over the internet	Start-ups, variable workloads, lowest cost
Private	Dedicated to one organisation (on-premise or hosted); maximum control and isolation	Highly sensitive or regulated data – banks, defence, healthcare
Hybrid	Private/on-premise + public cloud connected securely (VPN or dedicated link)	Keep sensitive data private, burst to public cloud for scale
Community	Shared by organisations with common requirements	Government departments, research institutions
Multi-cloud	Services from two or more public providers	Avoid vendor lock-in, use best-of-breed services

A4. AWS, Azure and Google Cloud – Service Mapping

Category	AWS	Microsoft Azure	Google Cloud
Virtual machines	EC2	Virtual Machines	Compute Engine
Serverless functions	Lambda	Azure Functions	Cloud Run functions
Object storage	S3	Blob Storage	Cloud Storage
Block storage	EBS	Managed Disks	Persistent Disk
File storage	EFS	Azure Files	Filestore
Archive storage	S3 Glacier	Archive tier	Archive storage class
Relational database	RDS, Aurora	Azure SQL Database	Cloud SQL, Spanner
NoSQL database	DynamoDB	Cosmos DB	Firestore, Bigtable
Kubernetes	EKS	AKS	GKE
Private network	VPC	Virtual Network (VNet)	VPC
Dedicated link	Direct Connect	ExpressRoute	Cloud Interconnect
Load balancing	Elastic Load Balancing	Load Balancer / Application Gateway	Cloud Load Balancing
CDN	CloudFront	Azure Front Door / CDN	Cloud CDN

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Category	AWS	Microsoft Azure	Google Cloud
DNS	Route 53	Azure DNS	Cloud DNS
Identity	IAM	Microsoft Entra ID	Cloud IAM
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring
Infrastructure as Code	CloudFormation	ARM templates / Bicep	Deployment Manager (Terraform works on all)
Message queue	SQS / SNS	Service Bus / Queue Storage	Pub/Sub

A5. Virtualization and Containers

- **Virtualization** lets one physical server run many isolated **virtual machines (VMs)**, each with its own guest OS, managed by a **hypervisor**. Type 1 (bare-metal: VMware ESXi, Microsoft Hyper-V, KVM) runs directly on hardware; Type 2 (hosted: VirtualBox, VMware Workstation) runs on a host OS.
- **Containers** package an application with its libraries and dependencies and run as isolated processes that **share the host OS kernel**. The kernel enforces isolation using **namespaces** (what a container can see) and **cgroups** (how much CPU, memory and which devices it can use).

Aspect	Virtual Machine	Container
Includes	Full guest OS + app	App + dependencies only
Size / start-up	GBs / minutes	MBs / seconds
Isolation	Stronger (separate kernels)	Lighter (shared kernel)
Different OSs on one host	Yes	No – same kernel family
Best for	Legacy apps, strong multi-tenant isolation	Microservices, CI/CD, portable deployments

Docker / Kubernetes term	Meaning
Dockerfile	Text instructions to build an image
Image	Read-only template of an application and its dependencies
Container	Running instance of an image
Registry	Stores and versions images – Docker Hub, Amazon ECR, Azure Container Registry
Pod	Smallest deployable unit in Kubernetes – one or more containers sharing network and storage
Node / Cluster	Worker machine / group of nodes managed by the control plane
Deployment / ReplicaSet	Declares desired number of pod replicas; supports rolling updates and rollback
Service	Stable network endpoint and load balancing for a set of pods
Horizontal Pod Autoscaler	Adds or removes pods based on CPU, memory or custom metrics
Self-healing	Failed containers are restarted and unhealthy pods replaced automatically

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A6. Cloud Storage

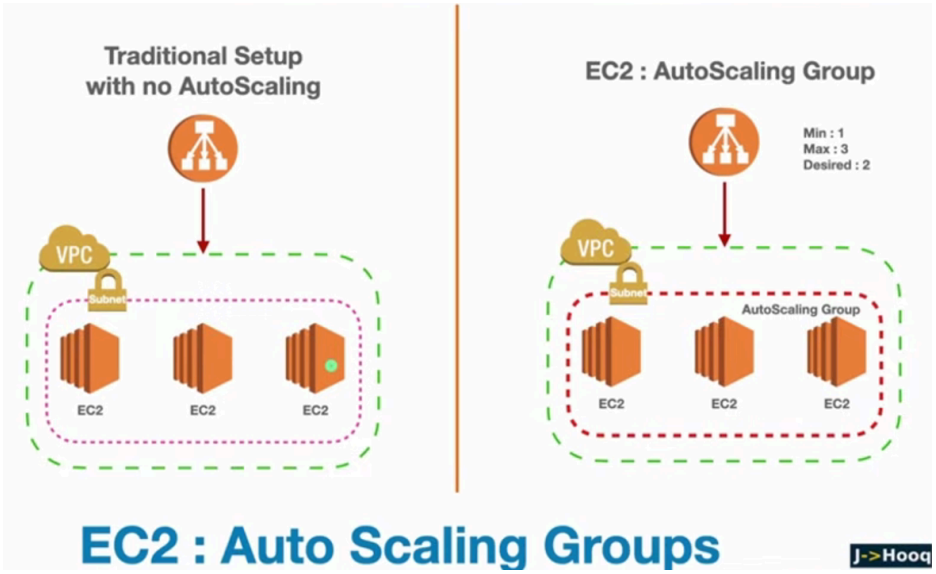
Type	How it works	Typical use	Examples
Block	Raw volumes attached to one VM; lowest latency	Databases, OS boot disks	EBS, Azure Managed Disks
File	Shared hierarchical file system over NFS/SMB	Shared folders, lift-and-shift apps	EFS, Azure Files
Object	Flat namespace of objects with metadata, accessed via HTTP API; virtually unlimited and scales automatically	Images, videos, backups, data lakes, static websites	S3, Azure Blob, Cloud Storage
Archive	Lowest cost, retrieval in minutes to hours	Compliance archives, long-term backups	S3 Glacier, Azure Archive tier

- **Storage tiers:** hot (frequent access), cool/infrequent, archive. **Lifecycle policies** move objects automatically between tiers by age.
- **Durability vs availability:** durability = probability data is not lost (S3 is designed for 99.999999999%, 'eleven nines'); availability = probability data can be accessed when needed.
- **Snapshot** = point-in-time copy of a volume; **replication** = continuous copy to another zone/region; **backup** = copy kept separately for restore.

A7. Scalability, Availability and Disaster Recovery

Term	Meaning
Vertical scaling (scale up)	Bigger machine – more CPU/RAM; limited and usually needs a restart
Horizontal scaling (scale out)	More machines behind a load balancer; needs stateless application design
Elasticity / Auto Scaling	Capacity added and removed automatically based on metrics or schedules
Load balancer	Distributes traffic; Layer 4 (TCP/UDP) or Layer 7 (HTTP – path/host-based routing); health checks remove failed servers
Region / Availability Zone	Region = geographic area; AZ = one or more isolated data centres within a region. Multi-AZ survives a data-centre failure; multi-region survives a regional disaster
Edge location / CDN	Caches content close to users for low latency
High availability vs fault tolerance	HA minimises downtime with quick failover; fault tolerance continues with zero interruption (full redundancy)
RPO / RTO	RPO = maximum acceptable data loss measured in time; RTO = maximum acceptable time to restore service

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DR strategy	Cost	Recovery speed
Backup & restore	Lowest	Slowest (hours)
Pilot light (core services minimal, scaled up on disaster)	Low	Tens of minutes
Warm standby (scaled-down full copy running)	Medium	Minutes
Multi-site active-active	Highest	Near zero RTO and RPO

SLA	Max downtime per year	Per month (≈30 days)
99%	3.65 days	7.2 hours
99.9%	8.76 hours	43.2 minutes
99.95%	4.38 hours	21.6 minutes
99.99%	52.56 minutes	4.32 minutes
99.999%	5.26 minutes	26 seconds

Composite availability: components in **series** multiply (99.9% × 99.9% = 99.8%); **redundant** components in parallel: $1 - (1 - A)^2$ (two 99% instances = 99.99%).

A8. Cloud Networking

Component	Purpose
VPC / VNet	Logically isolated private network in the cloud with your own IP range (CIDR)
Public subnet	Has a route to an Internet Gateway – for load balancers, bastion hosts
Private subnet	No direct internet route – for app servers and databases
Internet Gateway	Enables two-way internet access for public subnets
NAT Gateway	Lets private-subnet instances reach the internet (e.g., for patches) without being reachable from it
Route table	Rules that decide where traffic from a subnet goes

Component	Purpose
Security Group	Virtual firewall at instance level; STATEFUL (return traffic allowed automatically); allow rules only
Network ACL	Firewall at subnet level; STATELESS (inbound and outbound rules both needed); allow AND deny rules, evaluated in order
VPC Peering	Private connection between two VPCs; not transitive
Site-to-site VPN	Encrypted IPsec tunnel between on-premise network and VPC over the internet
Remote-access VPN	Individual users connect securely to the corporate/cloud network
Direct Connect / ExpressRoute	Dedicated private line – consistent bandwidth, does not use the public internet
API Gateway	Single managed entry point for APIs – routing, authentication, throttling

TCP connection basics (tested alongside cloud networking)

Step	Packet	Meaning
Open – 1	Client → SYN	Request to open a reliable connection, with an initial sequence number
Open – 2	Server → SYN-ACK	Server is available, accepts and sends its own sequence number (server state: SYN-RECEIVED)
Open – 3	Client → ACK	Connection ESTABLISHED – data transfer begins
Close	FIN → ACK, FIN → ACK	Graceful four-step termination
Refuse / abort	RST	Port closed or connection reset immediately

UDP has no handshake. A SYN flood sends many SYNs without completing the handshake to exhaust server resources. Cloud load-balancer TCP health checks pass when the handshake succeeds on the configured port.

A9. Cloud Security

Area	Key points
IAM	Users, groups, roles, policies; least privilege; MFA; roles for applications instead of stored access keys; regular access reviews
Encryption	At rest (disk, database, object encryption) and in transit (TLS); keys managed in a Key Management Service (AWS KMS, Azure Key Vault) with rotation
Secrets	Passwords, API keys and certificates stored in a secrets manager – never in code or Git
Perimeter & app protection	Security groups, NACLs, Web Application Firewall (blocks SQL injection, XSS), DDoS protection services
Common risks	Misconfiguration (public storage buckets, open ports), leaked credentials, insecure APIs, excessive permissions, insider threats
Zero Trust	Never trust, always verify – authenticate and authorise every request regardless of network location
Compliance	GDPR (EU personal data), HIPAA (US health data), PCI DSS (card payments), ISO 27001, SOC 2; India's DPDP Act 2023; data residency = keeping data in a specific country/region
Monitoring & audit	Logging of all API calls (CloudTrail, Azure Activity Log), alerts, security posture tools

A10. Serverless and Cloud-Native Design

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- **Serverless (FaaS):** event-driven, stateless functions; automatic scaling to zero; pay per execution; limits on execution time (AWS Lambda maximum 15 minutes); **cold start** = extra latency when a function runs after being idle.
- **Microservices:** small, independently deployable services communicating via APIs; each can scale and be updated separately. **Monolith:** one deployable unit – simpler but harder to scale parts independently.
- **Decoupling:** message queues (SQS, Service Bus) and pub/sub let producers and consumers work at different speeds without losing requests.
- **Caching:** in-memory caches (Redis, Memcached) reduce database load; **read replicas** scale read-heavy databases.
- **Stateless design:** keep sessions in a shared store so any server can handle any request – required for horizontal scaling.
- **Deployment strategies:** rolling (replace instances gradually), blue-green (switch traffic between two identical environments, instant rollback), canary (release to a small % of users first).

A11. Cost Management, DevOps and Operations

Topic	Key points
On-demand	Pay per second/hour, no commitment – unpredictable or short-term workloads
Reserved / Savings Plans	1–3 year commitment for large discounts – steady 24×7 workloads
Spot / Preemptible	Spare capacity at the deepest discounts but can be interrupted – fault-tolerant batch jobs
Cost control	Tagging, budgets and alerts, right-sizing, auto-shutdown of idle resources, storage lifecycle policies (FinOps)
Infrastructure as Code	Terraform, CloudFormation, ARM/Bicep – repeatable, version-controlled environments
Configuration management	Ansible, Chef, Puppet
CI/CD	Jenkins, GitHub Actions, Azure DevOps, AWS CodePipeline; artifacts and container images stored in registries
Observability	Metrics, logs and traces – CloudWatch, Azure Monitor, Prometheus, Grafana; troubleshoot latency from data before restarting or scaling

A12. Cloud Migration – the 7 Rs

Strategy	Meaning	Example
Rehost (lift and shift)	Move as-is to cloud VMs	On-premise app server copied to EC2
Replatform (lift and reshape)	Small optimisations, no code redesign	Self-managed database moved to managed RDS
Refactor / Re-architect	Redesign to be cloud-native	Monolith rebuilt as containerised microservices
Repurchase	Replace with a SaaS product	On-premise mail server replaced by Microsoft 365
Retire	Switch off applications no longer needed	Decommission an unused reporting tool
Retain	Keep on-premise for now	Mainframe kept due to regulation
Relocate	Move a virtualised environment to cloud at hypervisor level	VMware workloads moved to VMware Cloud on AWS

A13. Emerging Trends

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- **Edge computing:** processing data near its source (factories, vehicles, IoT) for low latency and reduced bandwidth.
- **Multi-cloud and hybrid management:** Azure Arc, Google Anthos, Kubernetes for portability.
- **AI/ML cloud services:** managed model training and generative AI platforms (Amazon Bedrock, Azure OpenAI Service, Google Vertex AI).
- **Green cloud:** carbon-aware regions and efficiency as part of sustainability goals.

Part B – MCQ Bank (184 Questions)

B1. Level 1 – Basics (25 Questions)

Q1. [P] Which is NOT one of NIST's five essential characteristics of cloud computing?

(A) On-demand self-service
(B) Rapid elasticity
(C) Measured service
(D)] Fixed long-term hardware contracts

Q2. [P] 'Pay only for what you use' reflects which characteristic?

(A) Resource pooling
[B] Measured service
(C) Broad network access
(D) Multi-tenancy

Q3. [P] Gmail is an example of:

(A) IaaS
(B) PaaS
(C)] SaaS
(D) FaaS

Q4. [P] Amazon EC2 is an example of:

(A)] IaaS
(B) PaaS
(C) SaaS
(D) DBaaS

Q5. [P] Azure App Service and Google App Engine are examples of:

(A) IaaS
(B)] PaaS
(C) SaaS
(D) On-premise

Q6. [P] A cloud owned by a provider and shared by many organisations over the internet is:

(A) Private
(B)] Public
(C) Hybrid
(D) Community

Q7. [P] Combining on-premise infrastructure with public cloud services is called:

(A) Multi-cloud
(B)] Hybrid cloud
(C) Community cloud
(D) Edge cloud

Q8. [P] Several government departments share a cloud built for their common security needs. This is a:

(A) Public cloud

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(B)] Community cloud
(C) Hybrid cloud
(D) Personal cloud

Q9. [P] Software that creates and runs virtual machines is a:

(A) Compiler
(B)] Hypervisor
(C) Container runtime
(D) Load balancer

Q10. [P] Amazon S3 is which type of storage?

(A) Block
(B) File
©] Object
(D) Tape

Q11. [P] Adding more servers to share the load is:

(A) Vertical scaling
(B)] Horizontal scaling
(C) Virtualization
(D) Replication

Q12. [P] Upgrading an existing server from 8 GB to 32 GB RAM is:

(A) Horizontal scaling
(B)] Vertical scaling
(C) Load balancing
(D) Elasticity

Q13. [P] Automatically adding and removing resources as demand changes is called:

(A)] Elasticity
(B) Durability
(C) Latency
(D) Redundancy

Q14. [P] Which component distributes incoming traffic across multiple servers?

(A) Load balancer
(B)] Firewall
(C) Hypervisor
(D) DNS cache

Q15. [P] Which service delivers website images and videos from locations close to users?

(A) VPN
(B)] CDN
(C) NAT Gateway
(D) IAM

Q16. [P] Isolated data centres within one cloud region are called:

(A) Edge locations
(B)] Availability Zones
(C) Subnets
(D) Clusters

Q17. [P] Docker is a:

(A) Hypervisor
(B)] Containerization platform
(C) Database
(D) Programming language

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Q18. [P] Kubernetes is mainly used for:

(A) Writing code
(B)] Container orchestration
(C) Database backup
(D) Email delivery

Q19. [P] Running code without provisioning or managing servers is called:

(A) Bare metal
(B)] Serverless computing
(C) Colocation
(D) Mainframe

Q20. [P] AWS Lambda and Azure Functions are examples of:

(A) IaaS
(B) SaaS
©] FaaS
(D) DaaS

Q21. [P] A logically isolated private network inside AWS is a:

(A)] VPC
(B) VPN
(C) CDN
(D) WAN

Q22. [P] Moving an application to cloud VMs without any code changes is called:

(A) Refactor
(B)] Rehost (lift and shift)
(C) Repurchase
(D) Retire

Q23. [P] Cloud adoption mainly changes IT spending from:

(A) OpEx to CapEx
(B)] CapEx to OpEx
(C) Fixed to capital
(D) Opex to taxes

Q24. [P] IAM stands for:

(A) Internet Access Management
(B)] Identity and Access Management
(C) Integrated Application Module
(D) Instance Allocation Manager

Q25. [P] Multi-factor authentication improves security by:

(A) Encrypting disks
(B)] Requiring an additional verification factor beyond the password
(C) Blocking all logins
(D) Creating backups

Answer Key with Explanations

Q	Ans	Explanation
1	D	The five are on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service.
2	B	Usage is metered and billed.

Q	Ans	Explanation
3	C	Users consume complete software.
4	A	Virtual machines where you manage the OS.
5	B	You deploy code; the platform manages servers and runtime.
6	B	Public cloud uses shared infrastructure.
7	B	Multi-cloud means more than one public provider.
8	B	Community clouds serve organisations with shared requirements.
9	B	Examples: ESXi, Hyper-V, KVM, VirtualBox.
10	C	Objects with metadata accessed via HTTP API.
11	B	Scale out.
12	B	Scale up.
13	A	Implemented with Auto Scaling.
14	A	Health checks remove failed servers from rotation.
15	B	CloudFront, Azure Front Door, Cloud CDN.
16	B	Spreading across AZs protects against a data-centre failure.
17	B	Builds and runs containers.
18	B	Schedules, scales and heals containers.
19	B	Provider runs and scales the code on demand.
20	C	Function as a Service.
21	A	Azure calls it a VNet.
22	B	Fastest migration path.
23	B	Upfront hardware purchases become operating expenses.
24	B	Controls who can do what on which resources.
25	B	E.g., password + OTP from an authenticator app.

B2. Level 2 – Intermediate (25 Questions)

Q1. [P] In IaaS, who is responsible for patching the guest operating system?

(A) Cloud provider
(B)] Customer
(C) Hardware vendor
(D) Nobody

Q2. [P] In SaaS, the customer is still responsible for:

(A) Server hardware
(B) Application code
©] Their data and user access
(D) Hypervisor patches

Q3. [P] Which cloud firewall is STATEFUL and works at the instance level?

(A) Network ACL
(B)] Security Group
(C) Route table
(D) Internet Gateway

Q4. [P] Which statement about Network ACLs is TRUE?

(A) They are stateful
(B) They support only allow rules
©] They are stateless, subnet-level and support allow and deny rules
(D) They attach to individual instances

Q5. [P] Instances in a private subnet must download OS updates but must not be reachable from the internet. Use:

(A) Internet Gateway on the private subnet
(B)] NAT Gateway
(C) VPC peering
(D) Elastic IP on every instance

Q6. [P] A subnet becomes 'public' when its route table has a route to:

(A) A NAT Gateway
(B)] An Internet Gateway
(C) A VPN
(D) Another subnet

Q7. [P] An SLA of 99.9% allows a maximum yearly downtime of about:

(A) 52.6 minutes
(B)] 8.76 hours
(C) 3.65 days
(D) 5.26 minutes

Q8. [P] An SLA of 99.99% allows a maximum yearly downtime of about:

(A)] 52.6 minutes
(B) 8.76 hours
(C) 4.38 hours
(D) 26 seconds

Q9. [P] A business says it can afford to lose at most 15 minutes of data. This value is the:

(A) RTO
(B)] RPO
(C) SLA
(D) TTL

Q10. [P] Which disaster-recovery strategy is the cheapest but slowest to recover?

(A) Multi-site active-active
(B) Warm standby
(C) Pilot light
(D)] Backup and restore

Q11. [A] Containers are lightweight because they:

(A) Include a full guest OS
(B)] Share the host operating-system kernel
(C) Run only on bare metal
(D) Need a Type 2 hypervisor

Q12. [P] In Docker, an image is to a container as:

(A) A running process is to a program
(B) A template is to a running instance
(C) A VM is to a hypervisor
(D)] A file is to a folder

Q13. [P] The smallest deployable unit in Kubernetes is a:

(A) Node

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(B)] Pod
(C) Cluster
(D) Service

Q14. [P] Spot instances are best suited for:

(A) Production databases
(B)] Fault-tolerant batch jobs that can be interrupted
(C) Payment gateways
(D) Domain controllers

Q15. [P] Reserved instances or savings plans give the best value for:

(A) Unpredictable short tests
(B)] Steady workloads running 24×7 for 1–3 years
(C) One-hour demos
(D) Interruptible jobs

Q16. [P] Which tool is used for Infrastructure as Code across multiple clouds?

(A)] Terraform
(B) Wireshark
(C) Jira
(D) Photoshop

Q17. [P] The Azure equivalent of AWS CloudFormation is:

(A) Azure Monitor
(B)] ARM templates / Bicep
(C) Azure DevOps Boards
(D) Entra ID

Q18. [P] A VM's operating-system disk uses which storage type?

(A) Object
(B)] Block
(C) Archive
(D) CDN

Q19. [P] Several Linux VMs must mount the same shared folder simultaneously. Use:

(A) Block storage
(B)] File storage (EFS / Azure Files)
(C) Archive storage
(D) Local disk

Q20. [P] Archive storage tiers are characterised by:

(A) Highest cost, instant access
(B)] Lowest cost, retrieval in minutes to hours
(C) Only for databases
(D) No durability

Q21. [P] A serverless function responds slowly on the first request after being idle. This is called:

(A) Throttling
(B)] Cold start
(C) Deadlock
(D) Cache miss

Q22. [P] A Type 1 hypervisor:

(A) Runs on top of Windows
(B)] Runs directly on the hardware
(C) Is a container engine
(D) Is a browser plugin

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Q23. [P] An application built as many small, independently deployable services follows:

(A) Monolithic architecture
(B)] Microservices architecture
(C) Client-server only
(D) Mainframe architecture

Q24. [P] Which component lets an order service keep accepting requests even if the payment service is slow?

(A)] Message queue
(B) DNS
(C) NAT Gateway
(D) Hypervisor

Q25. [P] Granting each user only the permissions needed for their job is the principle of:

(A) Defence in depth
(B)] Least privilege
(C) Separation of duties
(D) Zero downtime

Answer Key with Explanations

Q	Ans	Explanation
1	B	The provider manages only up to the virtualization layer.
2	C	Security IN the cloud – data and identities – is always the customer's.
3	B	Return traffic is automatically allowed.
4	C	Rules are evaluated in number order.
5	B	NAT allows outbound-only internet access.
6	B	Typically 0.0.0.0/0 → Internet Gateway.
7	B	0.1% of 8,760 hours.
8	A	0.01% of 525,600 minutes.
9	B	RPO = acceptable data loss; RTO = acceptable downtime.
10	D	Active-active is the most expensive and fastest.
11	B	No separate OS per container.
12	B	Containers are running instances of images.
13	B	A pod holds one or more containers.
14	B	Deep discounts but capacity can be reclaimed.
15	B	Discount in exchange for commitment.
16	A	CloudFormation (AWS) and Bicep (Azure) are provider-specific.
17	B	Declarative templates for Azure resources.
18	B	EBS / Azure Managed Disks.
19	B	File storage supports shared access over NFS/SMB.
20	B	Used for long-term retention.
21	B	The platform initialises a new execution environment.
22	B	Bare-metal hypervisors offer better performance.
23	B	Each service scales and deploys independently.

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Q	Ans	Explanation
24	A	Queues decouple producers from consumers.
25	B	Reduces damage from compromised accounts.

B3. Level 3 – Accenture-Pattern Scenarios (30 Questions)

Q1. [A] A cloud application's data grows unpredictably, and storage must scale automatically without manual intervention or capacity planning. Which storage option fits best?

(A) Block storage
(B) File storage
©] Object storage
(D) Tape storage

Q2. [A] An organisation moving to the cloud must protect highly sensitive customer data with maximum control and isolation. Which deployment model is typically the most secure?

(A) Public cloud
(B)] Private cloud
(C) Community cloud
(D) Multi-cloud

Q3. [A] An enterprise combines on-premise data centres with public cloud services and must connect them securely while protecting data integrity in transit. What should it implement?

(A) Public key infrastructure only
(B)] Secure hybrid connectivity with a site-to-site VPN
(C) Serverless architecture
(D) Larger VM sizes

Q4. [A] A cloud provider must run multiple isolated operating-system environments on a single physical server. Which technology is primarily used?

(A)] Virtual machines (virtualization)
(B) Bare metal
(C) CDN
(D) DNS

Q5. [A] A team adopts containerisation to simplify deployment. What best characterises containers?

(A) Virtual computers each running a full OS under a hypervisor
(B)] Packaged applications with dependencies that share the host OS kernel
(C) Separate physical servers
(D) Network segments inside a VPC

Q6. [A] Containers using dedicated hardware peripherals must be strictly isolated so each sees only its allocated devices. The most critical factor is:

(A) Equal CPU scheduling for all containers
(B)] Kernel-level resource isolation (cgroups/namespaces) limiting device access per container
(C) Sharing network interfaces between containers
(D) Adding more memory dynamically

Q7. [A] A containerised application shows high latency right after deployment. What should the DevOps engineer do FIRST?

(A) Restart all containers
(B) Reduce the number of containers
©] Analyse metrics, logs and network traffic to find the cause
(D) Increase CPU on every node

Q8. [A] Successful builds produce Docker images that must be stored and versioned so only tested images are deployed to staging and production. Use:

(A) Source code repository
(B)] Container registry / artifact repository
(C) Load balancer
(D) CDN

Q9. [A] An employee working from home needs secure access to the company's internal cloud applications. Which VPN fits?

(A) Site-to-site VPN
(B)] Remote-access VPN
(C) MPLS VPN
(D) VPC peering

Q10. [A] A company's on-premise office network must connect permanently to its cloud VPC so all office users can reach cloud servers. Use:

(A) Remote-access VPN on every laptop
(B)] Site-to-site VPN
(C) Public IPs on all servers
(D) CDN

Q11. [P] A start-up's API gets very irregular traffic; the team wants zero server management and to pay only per request. Best architecture:

(A) Reserved VMs
(B)] API Gateway + serverless functions
(C) Bare-metal servers
(D) On-premise cluster

Q12. [P] An e-commerce site expects a 10× traffic spike during a festival sale. Best approach:

(A) Buy bigger servers permanently
(B)] Auto Scaling group behind a load balancer
(C) Single large VM
(D) Manual server addition at midnight

Q13. [P] Users in Europe see slow image loads from an application hosted in Mumbai. The quickest improvement:

(A) Move the database
(B)] Use a CDN to cache images at edge locations
(C) Increase VM RAM
(D) Add a NAT Gateway

Q14. [P] An application must keep running even if an entire data centre fails. Deploy it:

(A) On one large VM
(B)] Across multiple Availability Zones behind a load balancer
(C) On a single AZ with backups
(D) On a developer laptop

Q15. [P] A bank must continue operating even if an entire cloud region becomes unavailable. Required design:

(A) Multi-AZ only
(B)] Multi-region disaster recovery
(C) Larger instances
(D) More security groups

Q16. [P] A developer accidentally pushed cloud access keys to a public GitHub repository. Immediate action:

(A)] Delete the repository only
(B) Revoke/rotate the keys immediately and move secrets to a secrets manager / IAM roles

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(C) Change the account name
(D) Ignore if repository is new

Q17. [P] A data leak occurred because a storage bucket was publicly readable. The root cause category is:

(A) DDoS
(B)] Misconfiguration
(C) Hardware failure
(D) Cold start

Q18. [P] A financial company must keep customer data inside India for regulatory reasons. It should:

(A) Use any region
(B)] Deploy in Indian cloud regions to meet data residency
(C) Use only a CDN
(D) Encrypt and store in any country

Q19. [P] A nightly 2-hour batch job can restart safely if interrupted, and cost must be minimum. Use:

(A) On-demand instances
(B) Reserved instances
©] Spot instances
(D) Dedicated hosts

Q20. [P] A production database server runs 24×7 and will be needed for three years. Lowest-cost option:

(A) Spot
(B) On-demand
©] Reserved instances / savings plan
(D) Serverless

Q21. [P] Teams need identical dev, test and production environments created repeatedly without manual clicks. Use:

(A)] Infrastructure as Code (Terraform)
(B) Screenshots of settings
(C) Email instructions
(D) Manual console setup

Q22. [P] A company replaces its on-premise email server with Microsoft 365. Which migration strategy is this?

(A) Rehost
(B) Replatform
©] Repurchase
(D) Retain

Q23. [P] An application is moved to the cloud and its self-managed database is switched to a managed database service without code redesign. This is:

(A) Rehost
(B)] Replatform
(C) Refactor
(D) Retire

Q24. [P] A monolithic application is rebuilt as containerised microservices. This is:

(A) Rehost
(B) Repurchase
[C] Refactor / Re-architect
(D) Relocate

Q25. [P] A web application on a VM must read files from object storage without storing access keys on the server. Use:

(A) Hard-coded keys
(B)] An IAM role attached to the VM

(C) Public bucket
(D) Shared password

Q26. [P] In a Kubernetes cluster, a crashed container is automatically restarted. This capability is:

(A)] Self-healing
(B) Cold start
(C) Peering
(D) Sharding

Q27. [P] Orders must not be lost when the payment service is temporarily overloaded. Best design:

(A) Direct synchronous calls only
(B)] Put orders in a message queue processed by the payment service
(C) Bigger payment server only
(D) Disable orders

Q28. [P] Students' assignment files are accessed often for 30 days and rarely after one year. Cost-efficient approach:

(A) Keep all in hot storage
(B)] Lifecycle policy moving files to cool and then archive tiers
(C) Delete after 30 days
(D) Store on VM disks

Q29. [P] A team needs only to deploy code; the provider should manage OS, patching and runtime. Choose:

(A) IaaS
(B)] PaaS
(C) On-premise
(D) Bare metal

Q30. [P] Who is responsible if a company's cloud database is breached because the admin password was 'admin123'?

(A) Cloud provider
(B)] Customer
(C) Internet provider
(D) Hardware vendor

Answer Key with Explanations

Q	Ans	Explanation
1	C	Object storage scales virtually without limit.
2	B	Dedicated infrastructure and full control.
3	B	An encrypted IPsec tunnel (or dedicated link) joins the environments.
4	A	Hypervisors create isolated VMs on one host.
5	B	Portable from developer laptop to production.
6	B	Isolation is enforced by the shared kernel.
7	C	Diagnose before acting.
8	B	ECR, ACR, Docker Hub, Artifactory.
9	B	Connects an individual device to the network.
10	B	Gateway-to-gateway tunnel for the whole network.
11	B	Scales to zero when idle.
12	B	Capacity grows and shrinks with demand.

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Q	Ans	Explanation
13	B	Content is served from locations near users.
14	B	Multi-AZ protects against a data-centre outage.
15	B	Regions are geographically separate.
16	B	Bots scan public repos for keys within minutes.
17	B	Block public access and apply least-privilege policies.
18	B	Data residency = data stays in a specified geography.
19	C	Interruptible workloads suit spot pricing.
20	C	Long-term commitment gives the biggest steady-state discount.
21	A	Version-controlled, repeatable infrastructure.
22	C	Moving to a SaaS product.
23	B	Lift and reshape.
24	C	Code and architecture are redesigned.
25	B	Roles provide temporary credentials automatically.
26	A	The control plane maintains the desired state.
27	B	Queues buffer requests until consumers catch up.
28	B	Automatic tiering by age.
29	B	PaaS abstracts the platform.
30	B	Identity and access are always the customer's responsibility.

B4. Level 4 – Advanced (20 Questions)

Q1. [P] A specific malicious IP address must be blocked for an entire subnet. Use:

(A)] Security group deny rule
(B) Network ACL deny rule
(C) Route table
(D) NAT Gateway

Q2. [P] Where must a NAT Gateway be placed so private instances can reach the internet?

(A) In the private subnet
(B)] In a public subnet, with the private route table pointing to it
(C) In another region
(D) Inside each instance

Q3. [P] VPC-A is peered with VPC-B, and VPC-B with VPC-C. Can VPC-A reach VPC-C through VPC-B?

(A) Yes, peering is transitive
(B)] No, peering is not transitive
(C) Only over IPv6
(D) Only with a CDN

Q4. [P] A company needs a private, consistent-bandwidth link to AWS that does not traverse the public internet. Use:

(A) Site-to-site VPN
(B)] AWS Direct Connect
(C) CloudFront
(D) Route 53

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Q5. [P] A web tier and a database are both required (in series), each with 99.9% availability. Combined availability is about:

(A) 99.9%
(B)] 99.8%
(C) 99.99%
(D) 98%

Q6. [P] Two redundant servers each have 99% availability; the service works if either is up. Combined availability:

(A) 98%
(B) 99%
©] 99.99%
(D) 99.5%

Q7. [P] A trading platform requires near-zero RTO and RPO. Which DR strategy?

(A) Backup and restore
(B) Pilot light
(C) Warm standby
(D)] Multi-site active-active

Q8. [P] Pods must scale automatically when average CPU exceeds 70%. Use:

(A)] Horizontal Pod Autoscaler
(B) Docker Compose
(C) Manual kubectl delete
(D) ConfigMap

Q9. [P] A release must switch all traffic to a new environment at once, with instant rollback to the old one. Use:

(A) Rolling update
(B)] Blue-green deployment
(C) Canary
(D) Recreate on the same servers

Q10. [P] A new feature is released to 5% of users first while metrics are watched. This is:

(A) Blue-green
(B)] Canary release
(C) Rolling back
(D) Hotfix

Q11. [P] An application stores user sessions on local server memory; after adding servers, users get logged out randomly. Fix:

(A) Use a bigger server
(B)] Store sessions in a shared cache/database so servers stay stateless
(C) Disable load balancer
(D) Use spot instances

Q12. [P] Encryption keys must be created, stored, rotated and audited centrally. Use:

(A)] Key Management Service (KMS / Key Vault)
(B) Spreadsheet
(C) Environment variables in code
(D) CDN

Q13. [P] 'Never trust, always verify every request, even inside the network' describes:

(A) Perimeter security
(B)] Zero Trust
(C) Defence in width
(D) Air gap

Q14. [P] A serverless function processing 2-hour video files keeps timing out. Best fix:

(A) Increase memory only
(B)] Move processing to containers or batch compute (or split into smaller steps)
(C) Use a CDN
(D) Add MFA

Q15. [P] A factory's IoT sensors need decisions within milliseconds without sending all data to a distant region. Use:

(A)] Edge computing
(B) Archive storage
(C) Multi-region DR
(D) SaaS

Q16. [P] A key business reason to adopt multi-cloud is:

(A) Lower complexity
(B)] Avoiding vendor lock-in and using best-of-breed services
(C) Fewer skills required
(D) No need for networking

Q17. [P] A Web Application Firewall mainly protects against:

(A) Power failures
(B)] Layer-7 attacks such as SQL injection and cross-site scripting
(C) Disk corruption
(D) Cold starts

Q18. [P] Auto Scaling added web servers, but the site is still slow because the database is overloaded by read queries. Best fix:

(A) Add more web servers
(B)] Add database read replicas and/or a cache
(C) Restart the load balancer
(D) Disable logging

Q19. [P] Frequently read product catalogue data should be served without hitting the database every time. Use:

(A)] In-memory cache (Redis / ElastiCache)
(B) Archive tier
(C) NAT Gateway
(D) VPN

Q20. [P] The three pillars of observability are:

(A) CPU, RAM, disk
(B)] Metrics, logs and traces
(C) IaaS, PaaS, SaaS
(D) Build, test, deploy

Answer Key with Explanations

Q	Ans	Explanation
1	B	Security groups support only allow rules; NACLs support deny.
2	B	NAT itself needs internet access via the Internet Gateway.
3	B	A direct peering or a transit gateway is required.
4	B	Azure equivalent: ExpressRoute.
5	B	0.999 × 0.999 = 0.998.
6	C	1 − (0.01 × 0.01) = 0.9999.

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Q	Ans	Explanation
7	D	Both sites serve traffic continuously.
8	A	HPA adjusts replica count based on metrics.
9	B	Two identical environments; flip traffic.
10	B	Gradual exposure limits impact.
11	B	Stateless design is essential for horizontal scaling.
12	A	Managed key services integrate with storage and databases.
13	B	Identity-centric security.
14	B	AWS Lambda has a 15-minute maximum execution time.
15	A	Processing near the data source.
16	B	It increases management complexity.
17	B	It inspects HTTP requests.
18	B	Scale the actual bottleneck.
19	A	Caches reduce latency and database load.
20	B	Together they explain what failed, where and why.

B5. Table-based Case Studies (12 Questions)

Case A – Three-tier Application SLAs

Tier	Service	Monthly SLA
Web	Managed load balancer + web servers	99.95%
Application	Container cluster	99.9%
Database	Managed relational database (multi-AZ)	99.99%

Case A Questions

Q1. All three tiers are required for the application to work. The composite availability is about:

(A) 99.99%
(B) 99.95%
©] 99.84%
(D) 99.9%

Q2. Which tier limits the overall availability the most?

(A) Web
(B)] Application
(C) Database
(D) All equal

Q3. Maximum downtime per year allowed by the database SLA alone is about:

(A) 5.3 minutes
(B)] 52.6 minutes
(C) 4.4 hours
(D) 8.8 hours

Q4. Approximate yearly downtime for the composite application:

(A) 1 hour
(B)] 14 hours

(C) 52 minutes
(D) 3 days

Answer Key with Explanations

Q	Ans	Explanation
1	C	0.9995 × 0.999 × 0.9999 ≈ 0.9984.
2	B	Lowest individual SLA (99.9%).
3	B	0.01% of 525,600 minutes.
4	B	0.16% of 8,760 hours ≈ 14 hours.

Case B – Choosing a Pricing Model

Option	Effective rate	Notes
On-demand	₹10 per hour	No commitment
Reserved (1 year)	₹6 per hour	Pay for all 8,760 hours of the year
Spot	₹3 per hour	Can be interrupted with short notice

Workload	Pattern
W1 – Production database	Runs 24×7 for the whole year
W2 – Report generation	2 hours every night, restartable if interrupted
W3 – Load test	Runs 24 hours a day for 14 days only, must not be interrupted

Case B Questions

Q1. Best pricing option for W1:

(A) Spot
(B) On-demand
©] Reserved
(D) Any

Q2. Yearly saving for W1 using Reserved instead of On-demand:

(A)] ₹35,040
(B) ₹52,560
(C) ₹87,600
(D) ₹17,520

Q3. Best pricing option for W2 and its yearly cost:

(A)] Spot, ₹2,190
(B) On-demand, ₹7,300
(C) Reserved, ₹52,560
(D) Spot, ₹7,300

Q4. Best option for W3 and its cost:

(A) Reserved, ₹52,560
(B)] On-demand, ₹3,360
(C) Spot, ₹1,008
(D) On-demand, ₹87,600

Answer Key with Explanations

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Q	Ans	Explanation
1	C	Steady 24×7 usage for a year.
2	A	(10 − 6) × 8,760 = ₹35,040.
3	A	2 h × 365 × ₹3 = ₹2,190.
4	B	Short-term and cannot be interrupted: 24 × 14 × ₹10 = ₹3,360. A one-year reservation would cost far more.

Case C – Architecture Requirements

#	Requirement
R1	Store millions of product images accessed over HTTP
R2	Run containerised microservices with managed Kubernetes
R3	Alert the team when CPU utilisation crosses 80%
R4	Route the domain www.shop.in to the application

Case C Questions

Q1. R1 is best met on AWS / Azure by:

(A) EBS / Managed Disks
(B)] S3 / Blob Storage
(C) EFS / Azure Files
(D) RDS / Azure SQL

Q2. R2 is best met on AWS / Azure by:

(A) EC2 / Virtual Machines
(B) Lambda / Functions
©] EKS / AKS
(D) Route 53 / Azure DNS

Q3. R3 is best met on AWS / Azure by:

(A)] CloudWatch / Azure Monitor alarms
(B) IAM / Entra ID
(C) CloudFront / Front Door
(D) Glacier / Archive tier

Q4. R4 is best met on AWS / Azure by:

(A)] Route 53 / Azure DNS
(B) VPC / VNet
(C) SQS / Service Bus
(D) KMS / Key Vault

Answer Key with Explanations

Q	Ans	Explanation
1	B	Object storage.
2	C	Managed Kubernetes services.
3	A	Metric alarms trigger notifications.
4	A	DNS services map domain names to endpoints.

B6. Accenture-Pattern Scenarios – Set 2 (30 Questions)

Q1. [A][L3] A retail chain wants transactions from 500 stores analysed within seconds to flag suspicious payments. Best approach:

(A) Nightly batch job on a VM
(B)] Real-time streaming ingestion with event-driven processing (e.g., Kinesis / Event Hubs + functions)
(C) Archive storage
(D) Weekly Excel reports

Q2. [A][L3] A healthcare app stores patient records in a cloud database and must control exactly who can decrypt them. Best approach:

(A) No encryption, rely on firewall
(B)] Encryption at rest with customer-managed keys in a Key Management Service and strict key access policies
(C) Store the key inside the application code
(D) Use a CDN

Q3. [A][L3] The monthly cloud bill doubled unexpectedly. What should the team do FIRST?

(A) Delete all resources
(B)] Analyse cost reports by service and tags to find the resources driving the increase
(C) Move to another provider
(D) Buy reserved instances immediately

Q4. [A][L3] Development VMs run 24×7 but are used only 9 AM–6 PM on weekdays. Most effective saving:

(A) Bigger VMs
(B)] Scheduled automatic stop/start outside working hours
(C) Spot for production
(D) Add more regions

Q5. [A][L3] An API hosted in one region is slow for users on other continents. Best improvement for dynamic API traffic:

(A) CDN for images only
(B)] Deploy in multiple regions with latency-based DNS routing / global load balancing
(C) Increase disk size
(D) Use archive storage

Q6. [A][L3] A start-up wants a relational database with automated backups, patching and failover, without managing servers. Use:

(A) Database installed on a VM
(B)] Managed database service (RDS / Azure SQL Database)
(C) Object storage
(D) Spreadsheet

Q7. [A][L3] A mobile app needs user sign-up, login and Google sign-in without building authentication from scratch. Use:

(A) Custom login code on a VM
(B) Managed identity service (Amazon Cognito / Microsoft Entra External ID)
©] VPN
(D) Message queue

Q8. [A][L3] A legacy application needs a specific Windows Server version and registry settings. Best cloud option:

(A) Serverless functions
(B)] Virtual machine (IaaS)
(C) SaaS
(D) Static website

Q9. [A][L3] A company website is plain HTML, CSS and JavaScript with no server-side code. Lowest-cost, highly available hosting:

(A) Two large VMs

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(B)] Object storage static website hosting with a CDN
(C) Kubernetes cluster
(D) Dedicated host

Q10. [A][L3] A security review finds database passwords hard-coded in application config files. Fix:
(A) Encrypt the config file with a ZIP password
(B)] Store credentials in a secrets manager with automatic rotation and grant access via IAM role
(C) Rename the file
(D) Make the repository private only

Q11. [A][L3] Configuration values must be changed without rebuilding container images in Kubernetes. Use:
(A) Hard-code values
(B)] ConfigMaps (and Secrets for sensitive values)
(C) Pods only
(D) Node labels

Q12. [A][L3] Two business units share one cloud account and need separate billing and permissions. Best practice:
(A) Share one admin password
(B)] Separate accounts / subscriptions under a central organisation with consolidated billing
(C) One giant resource group
(D) Disable IAM

Q13. [A][L3] Auditors ask who deleted a production storage bucket last week. Where is this answered?
(A) CDN logs
(B)] Cloud audit logs of API calls (CloudTrail / Azure Activity Log)
(C) Load balancer health checks
(D) DNS records

Q14. [A][L3] A VM's security group allows SSH (port 22) from 0.0.0.0/0. Best remediation:
(A) Leave it, password is strong
(B)] Restrict access to known IP ranges or use a bastion host / session manager
(C) Open all ports
(D) Change the VM name

Q15. [A][L3] 200 TB of archives must move to the cloud, but the office internet link is slow. Best option:
(A) Upload over the existing link
(B)] Offline transfer appliance (AWS Snowball / Azure Data Box)
(C) E-mail the files
(D) CDN

Q16. [A][L3] Users sometimes delete or overwrite important files in object storage. How can earlier copies be recovered?
(A)] Enable versioning (and soft delete)
(B) Increase bucket size
(C) Use spot instances
(D) Add MFA only

Q17. [A][L3] An app runs on a developer's laptop but fails in testing because of different library versions. The solution that ensures identical environments:
(A) Write longer documentation
(B)] Containerise the application with its dependencies
(C) Use a bigger test server
(D) Disable tests

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Q18. [A][L3] Analysts occasionally run SQL queries on terabytes of log files and want no infrastructure to manage. Use:

(A) Self-managed Hadoop cluster
(B)] Serverless query / data-warehouse service (Athena / BigQuery / Synapse serverless)
(C) VM with MySQL
(D) Excel

Q19. [A][L3] Servers behind a load balancer must be patched without downtime. Approach:

(A) Patch all servers together
(B)] Rolling update – patch one or a few at a time while others serve traffic
(C) Shut the site for maintenance
(D) Delete the load balancer

Q20. [A][L3] A website is flooded with traffic from thousands of compromised devices. Best defence:

(A) Bigger single server
(B)] DDoS protection service with WAF and CDN in front of the application
(C) Change the domain name
(D) Disable logging

Q21. [A][L3] The login page receives repeated SQL-injection attempts. Which control blocks them before they reach the app?

(A)] Web Application Firewall
(B) NAT Gateway
(C) Archive tier
(D) Auto Scaling

Q22. [A][L3] A regulator requires that the bank's servers do not share physical hardware with other customers. Use:

(A) Shared public VMs
(B)] Dedicated hosts / private cloud
(C) Spot instances
(D) Serverless

Q23. [A][L3] An 'order placed' event must notify the inventory, billing and e-mail services at the same time. Use:

(A) Single queue with one consumer
(B)] Publish/subscribe messaging (SNS / Event Grid / Pub/Sub)
(C) Direct database polling
(D) FTP

Q24. [A][L3] A manufacturer needs cloud-managed services but data must stay inside its own factory for latency reasons. Use:

(A) Public region only
(B)] Hybrid offerings that run cloud services on-premise (AWS Outposts / Azure Stack)
(C) CDN
(D) SaaS e-mail

Q25. [A][L3] One web server fails its health checks. What does the load balancer do?

(A) Sends more traffic to it
(B)] Stops routing traffic to it until it is healthy again
(C) Shuts down all servers
(D) Deletes the DNS record

Q26. [A][L3] A single-AZ database outage took the whole application down. Best fix:

(A) Bigger instance in the same AZ
(B)] Enable multi-AZ deployment with automatic failover
(C) Add a CDN

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(D) More web servers

Q27. [A][L3] An external vendor needs to download one file for only 24 hours. Most secure approach:

(A) Make the bucket public
(B)] Share a time-limited pre-signed URL / SAS token
(C) Give the vendor admin credentials
(D) E-mail the root password

Q28. [A][L3] Production containers run as root in privileged mode. Best practice is to:

(A) Keep root for convenience
(B)] Run as non-root with minimal privileges and no privileged mode
(C) Disable logging
(D) Use larger nodes

Q29. [A][L3] Microservices need consistent traffic routing, mutual TLS and request tracing without changing application code. Use:

(A)] Service mesh (e.g., Istio)
(B) FTP server
(C) Archive tier
(D) NAT Gateway

Q30. [A][L3] A customer-support team wants an AI chatbot quickly without training its own models. Best option:

(A) Build GPUs on-premise
(B)] Use managed AI / generative-AI cloud services
(C) Excel macros
(D) Bare-metal servers

Answer Key with Explanations

Q	Ans	Explanation
1	B	Streaming services process events as they arrive.
2	B	Customer-managed keys give control over decryption and key rotation.
3	B	Diagnose with Cost Explorer / Cost Management before acting.
4	B	Removes roughly 70% of idle hours.
5	B	Requests are served from the nearest healthy region.
6	B	Database as a Service handles routine administration.
7	B	Identity platforms provide secure authentication and social login.
8	B	IaaS gives full OS control.
9	B	No servers to manage; scales automatically.
10	B	Secrets must never live in code or configuration repositories.
11	B	Configuration is injected as environment variables or files.
12	B	Account boundaries isolate cost and access.
13	B	Every management API call is recorded with identity and time.
14	B	Open management ports are a top attack vector.
15	B	Physical devices ship large data volumes faster than slow networks.
16	A	Versioning keeps previous object versions.
17	B	Containers solve 'works on my machine' problems.

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Q	Ans	Explanation
18	B	Pay per query; no clusters to maintain.
19	B	Capacity stays available throughout.
20	B	Edge networks absorb and filter attack traffic.
21	A	WAF inspects HTTP requests for malicious patterns.
22	B	Single-tenant hardware meets isolation requirements.
23	B	Pub/sub fans out one event to many subscribers.
24	B	Cloud APIs and management, physically on-site.
25	B	Health checks enable automatic failover.
26	B	A standby replica in another AZ takes over.
27	B	Access expires automatically.
28	B	Reduces impact if a container is compromised.
29	A	Sidecar proxies handle networking concerns.
30	B	Pre-built cloud AI services shorten delivery time.

B7. Placement-Paper Pattern – TCS, Infosys, Capgemini, Cognizant, Wipro (20 Questions)

Q1. [P][L1] Which of the following is an example of PaaS?

(A) Google App Engine
(B) Gmail
(C) Amazon EC2
(D) Dropbox

Q2. [P][L1] Among IaaS, PaaS and SaaS, which gives the customer the MOST control?

(A) SaaS
(B) PaaS
(C) IaaS
(D) All equal

Q3. [P][L1] Which of the following is NOT a cloud service provider?

(A) AWS
(B) Microsoft Azure
(C) Google Cloud
(D) Apache Tomcat

Q4. [P][L1] SLA stands for:

(A) System Level Access
(B) Service Level Agreement
(C) Software License Agreement
(D) Secure Layer Architecture

Q5. [P][L1] Multi-tenancy in cloud computing means:

(A) One customer per server
(B) Many customers share the same infrastructure with logical isolation
(C) Multiple data centres
(D) Multiple passwords

Q6. [P][L1] Which is a container orchestration tool?

(A) Jenkins
(B) Git

(C) Kubernetes
(D) Nagios

Q7. [P][L1] Which hypervisor is a Type 1 (bare-metal) hypervisor?

(A) VMware ESXi
(B) Oracle VirtualBox
(C) VMware Workstation
(D) Parallels Desktop

Q8. [P][L1] Object storage is usually accessed using:

(A) SATA
(B) HTTP/HTTPS REST APIs
(C) Parallel ports
(D) SMTP

Q9. [P][L1] Which is a NoSQL database service?

(A) Amazon DynamoDB
(B) Azure SQL Database
(C) Amazon RDS for MySQL
(D) Cloud SQL

Q10. [P][L2] 'Cloud bursting' refers to:

(A) Deleting cloud data
(B) An application running on-premise overflowing into public cloud during peak demand
(C) A DDoS attack
(D) Data corruption

Q11. [P][L2] Vendor lock-in means:

(A) Data is encrypted
(B) Difficulty moving to another provider because of proprietary services
(C) Provider locks data centre doors
(D) Account suspension

Q12. [P][L2] Which is a disadvantage of public cloud?

(A) No upfront cost
(B) Less direct control and dependence on internet connectivity
(C) Rapid scaling
(D) Global reach

Q13. [P][L2] Elasticity differs from scalability because elasticity:

(A) Is only manual
(B) Automatically adds and removes resources to match short-term demand
(C) Only increases resources
(D) Applies only to storage

Q14. [P][L2] In SaaS, end users mainly interact with which layer?

(A) Hardware
(B) Hypervisor
(C) Application
(D) Operating system

Q15. [P][L2] Serverless functions are billed mainly on:

(A) Number of servers
(B) Number of executions and execution time/memory
(C) Monthly fixed fee only
(D) Disk size

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Q16. [P][L2] Replicating data to another region mainly improves:

(A) Typing speed
(B) Durability and disaster recovery
(C) Font rendering
(D) CPU clock speed

Q17. [P][L2] A key benefit of IaaS is:

(A) No need to buy or maintain physical servers
(B) No need to manage the OS
(C) No need for internet
(D) Free software licences

Q18. [P][L2] Which statement about hybrid cloud is correct?

(A) It uses only public cloud
(B) It combines private/on-premise and public cloud with connectivity between them
(C) It means two SaaS products
(D) It cannot use VPNs

Q19. [P][L2] Which service type does Microsoft 365 belong to?

(A) IaaS
(B) PaaS
(C) SaaS
(D) CaaS

Q20. [P][L2] Which cloud feature lets resources be provisioned through a web portal without human interaction with the provider?

(A) Measured service
(B) On-demand self-service
(C) Resource pooling
(D) Broad network access

Answer Key with Explanations

Q	Ans	Explanation
1	A	Gmail and Dropbox are SaaS; EC2 is IaaS.
2	C	The customer controls OS and everything above it.
3	D	Tomcat is a Java web server.
4	B	Defines guaranteed availability and support.
5	B	Enables resource pooling.
6	C	Jenkins = CI, Git = version control, Nagios = monitoring.
7	A	The others run on a host OS.
8	B	Objects are read and written via API calls.
9	A	The others are relational.
10	B	A typical hybrid-cloud pattern.
11	B	Multi-cloud and open standards reduce it.
12	B	The others are advantages.
13	B	Scalability is the ability to grow; elasticity is automatic grow-and-shrink.
14	C	Everything else is managed by the provider.

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Q	Ans	Explanation
15	B	No charge when idle.
16	B	Data survives a regional outage.
17	A	In IaaS the customer still manages the OS.
18	B	Workloads and data can move between environments.
19	C	Ready-to-use productivity software.
20	B	One of the five NIST characteristics.

B8. Recent-Drive Level – Deployment Models & TCP Connections (22 Questions)

Q1. [A][L3] As part of IT governance, you must choose a cloud deployment model for workloads holding highly sensitive customer data. Which model typically offers the strongest security and control?

(A) Public cloud
(B) Private cloud
(C) Hybrid cloud
(D) Community cloud

Q2. [A][L3] A company wants low-cost public hosting for its marketing website but must keep customer financial records in its own data centre. Best model:

(A) Public only
(B) Private only
(C) Hybrid cloud
(D) Community cloud

Q3. [A][L3] Several hospitals with the same regulatory requirements decide to share one cloud environment. This is a:

(A) Public cloud
(B) Community cloud
(C) Hybrid cloud
(D) Multi-cloud

Q4. [A][L3] A start-up handling no sensitive data needs the lowest upfront cost and fastest setup. Best model:

(A) Private cloud
(B) Public cloud
(C) On-premise data centre
(D) Community cloud

Q5. [A][L3] Which deployment model gives an organisation full control over physical access, security policies and hardware?

(A) Public
(B) Private
(C) Community
(D) SaaS

Q6. [A][L3] The main trade-off of choosing a private cloud is:

(A) Weaker security
(B) Higher cost and maintenance responsibility
(C) No virtualization
(D) No internet access possible

Q7. [A][L3] Before deciding which data can move to public cloud, an IT governance team should FIRST:

(A) Buy reserved instances
(B) Classify data by sensitivity and regulatory requirements

(C) Choose a CDN
(D) Migrate everything

Q8. [A][L3] The key security concern with storing highly sensitive data in a public cloud is:

(A) Too much control
(B) Shared multi-tenant infrastructure and less direct control
(C) No encryption available
(D) Cannot scale

Q9. [A][L2] A company runs compute on AWS and analytics on Google Cloud. This is:

(A) Hybrid cloud
(B) Multi-cloud
(C) Community cloud
(D) Private cloud

Q10. [A][L3] A defence organisation must meet strict national-security rules for classified data. Most appropriate:

(A) Public cloud shared tenancy
(B) Private (sovereign / on-premise) cloud
(C) Community cloud with any members
(D) Free SaaS tools

Q11. [A][L2] After a client application sends a SYN packet to open a reliable connection, a server that is available and ready responds with:

(A) FIN-ACK
(B) SYN-ACK
(C) RST
(D) ACK only

Q12. [A][L2] After receiving SYN-ACK, the client completes the handshake by sending:

(A) SYN
(B) FIN
(C) ACK
(D) RST

Q13. [A][L2] A client sends SYN to a port where no service is listening. The server responds with:

(A) SYN-ACK
(B) RST
(C) FIN
(D) No reply ever

Q14. [A][L2] A graceful TCP connection close is initiated with which flag?

(A) SYN
(B) RST
(C) FIN
(D) URG

Q15. [A][L1] How many segments are exchanged to establish a TCP connection?

(A) 2
(B) 3
(C) 4
(D) 5

Q16. [A][L2] How many segments are typically exchanged for graceful TCP termination?

(A) 2
(B) 3
(C) 4
(D) 1

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Q17. [A][L2] TCP sequence numbers are mainly used to:

(A) Encrypt data
(B) Order segments and detect lost or duplicate data
(C) Assign IP addresses
(D) Compress packets

Q18. [A][L3] An attacker sends thousands of SYN packets but never completes the handshake, exhausting server resources. This is a:

(A) SYN flood (DoS)
(B) Packet sniffing
(C) IP fragmentation
(D) DNS spoofing

Q19. [A][L2] Which transport protocol has NO connection-establishment handshake?

(A) TCP
(B) UDP
(C) HTTPS
(D) SSH

Q20. [A][L2] Which TCP flag confirms receipt of data?

(A) SYN
(B) ACK
(C) FIN
(D) PSH

Q21. [A][L3] A server has sent SYN-ACK but has not yet received the final ACK. Its connection state is:

(A) ESTABLISHED
(B) SYN-RECEIVED (half-open)
(C) TIME-WAIT
(D) CLOSED

Q22. [A][L3] A cloud load balancer uses a TCP health check on port 443. A target is marked healthy when:

(A) It responds to ping
(B) The TCP handshake on port 443 completes successfully
(C) Its CPU is below 50%
(D) DNS resolves its name

Answer Key with Explanations

Q	Ans	Explanation
1	B	Dedicated infrastructure, full control of security policies and access.
2	C	Sensitive data stays private while public workloads use public cloud.
3	B	Shared by organisations with common concerns.
4	B	Pay-as-you-go with no hardware investment.
5	B	The organisation owns or exclusively uses the infrastructure.
6	B	Control comes with capital and operational cost.
7	B	Data classification drives the deployment decision.
8	B	Strong controls exist, but infrastructure is shared.
9	B	Two or more public providers.
10	B	Maximum isolation and national control.

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Q	Ans	Explanation
11	B	Second step of the three-way handshake.
12	C	SYN → SYN-ACK → ACK.
13	B	RST refuses the connection.
14	C	RST aborts; FIN closes gracefully.
15	B	Three-way handshake.
16	C	FIN, ACK, FIN, ACK.
17	B	They make TCP reliable and ordered.
18	A	Half-open connections fill the server's queue.
19	B	UDP is connectionless.
20	B	Acknowledgement.
21	B	It becomes ESTABLISHED after the ACK arrives.
22	B	TCP health checks test whether the port accepts connections.

Talent Shine exam tips for cloud questions

Read the scenario's last line, then pick the KEY WORD: 'sensitive data' → private cloud; 'scales automatically / unlimited' → object storage; 'connect on-premise to cloud' → site-to-site VPN; 'home user' → remote-access VPN; 'isolated OS on one server' → VMs; 'lightweight, shares kernel, portable' → containers; 'no server management, pay per request' → serverless; 'first step' → diagnose with metrics and logs.

Classic traps: security group vs NACL, RPO vs RTO, multi-AZ vs multi-region, rehost vs replatform vs refactor, spot vs reserved, durability vs availability.

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